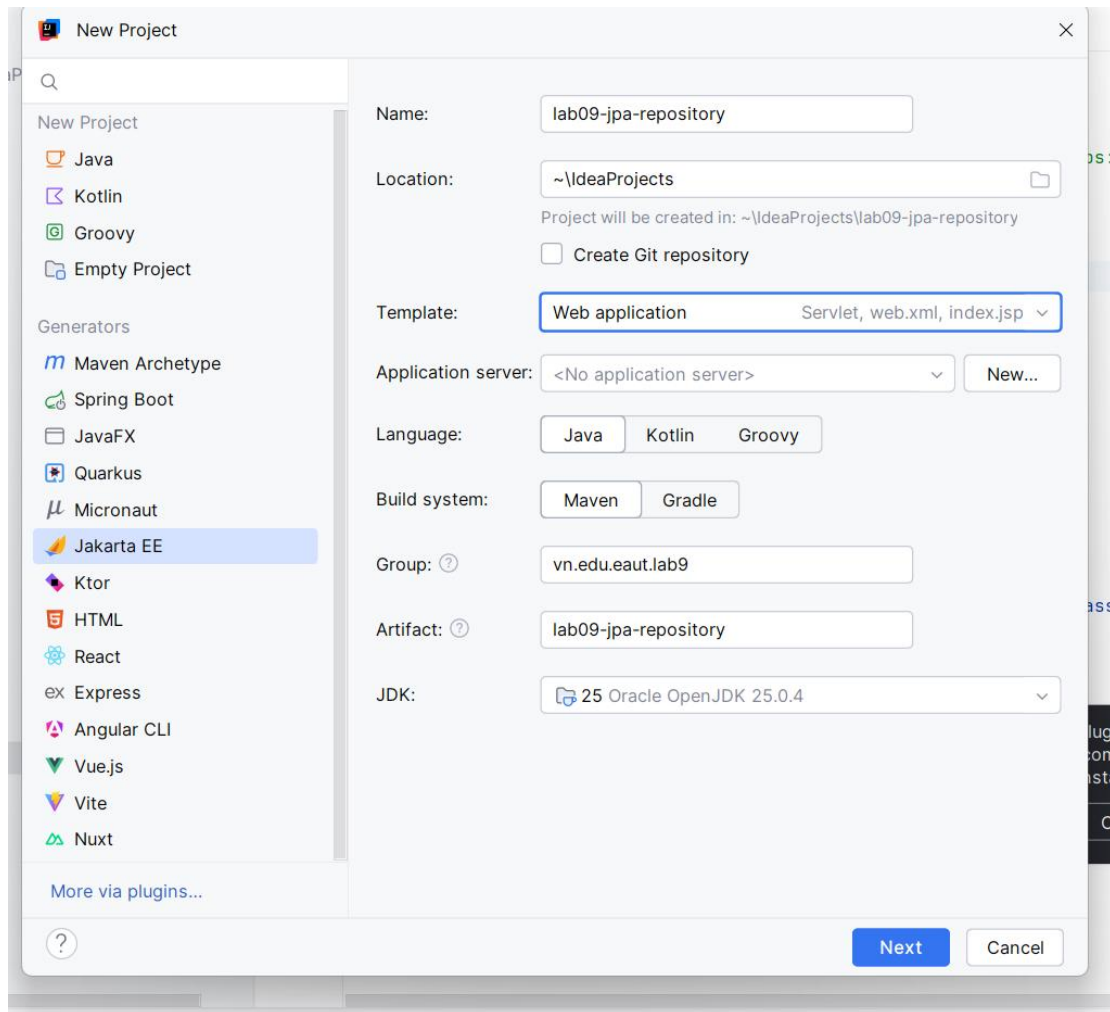


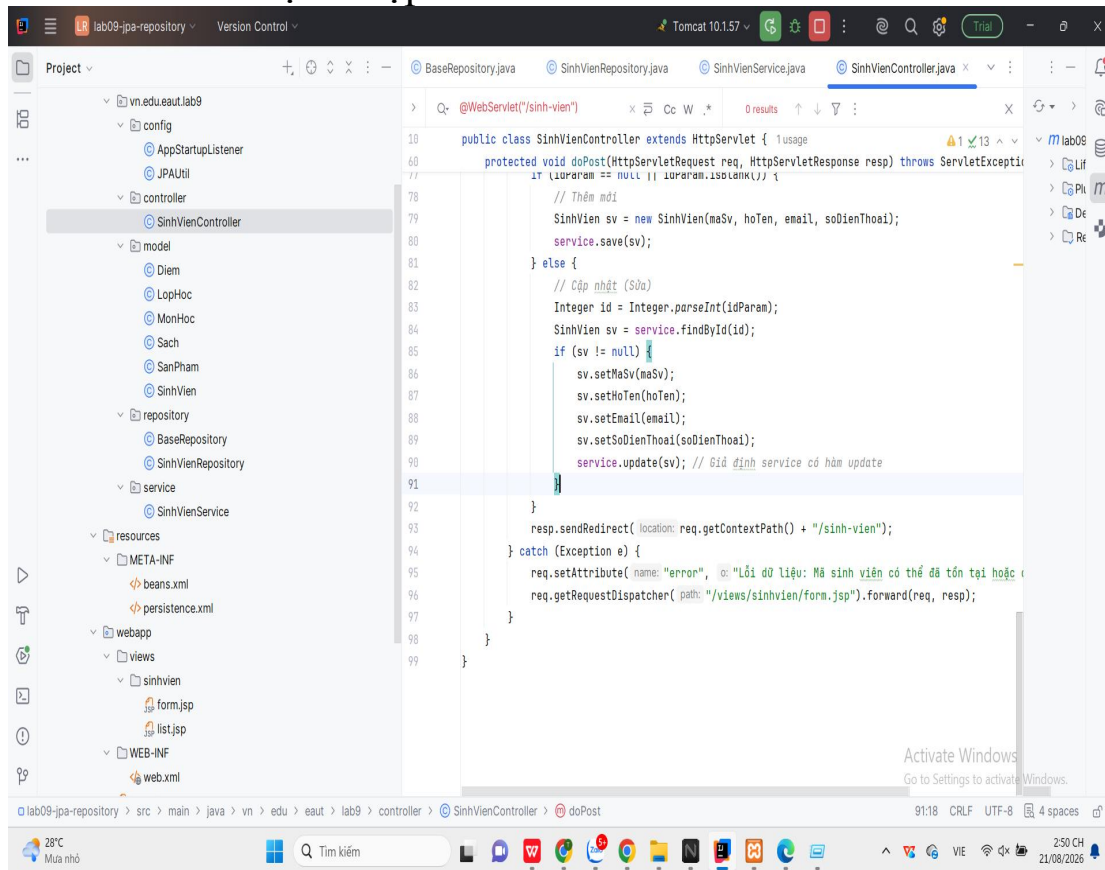
LAP 9

HỌ TÊN : TRẦN VIỆT ĐỨC

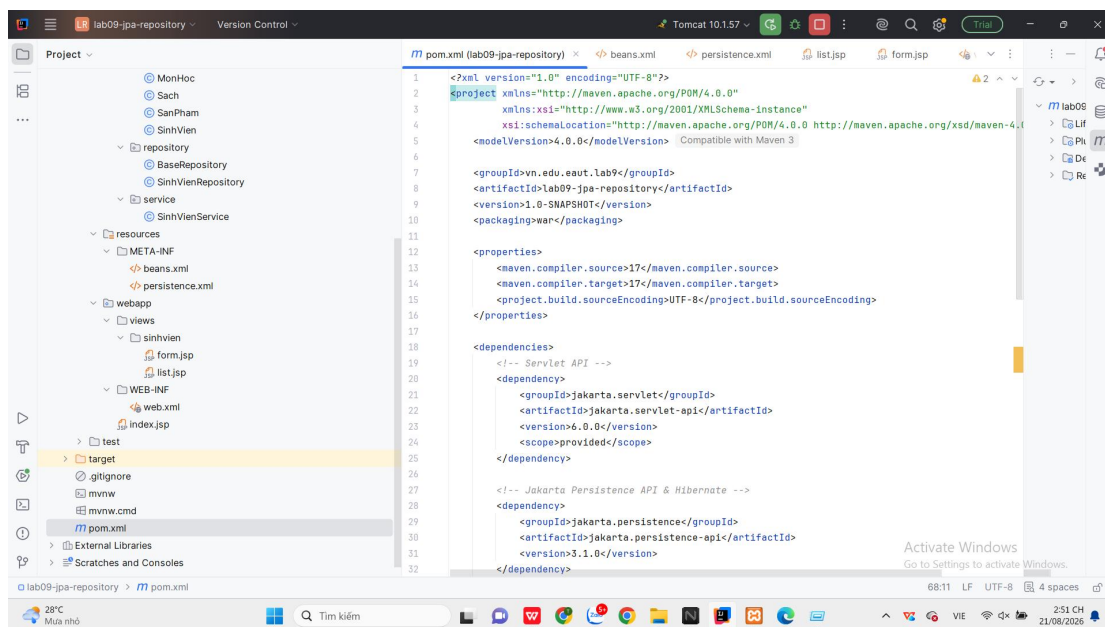
MSV : 20220111



Cầu hình toàn bộ bài tập

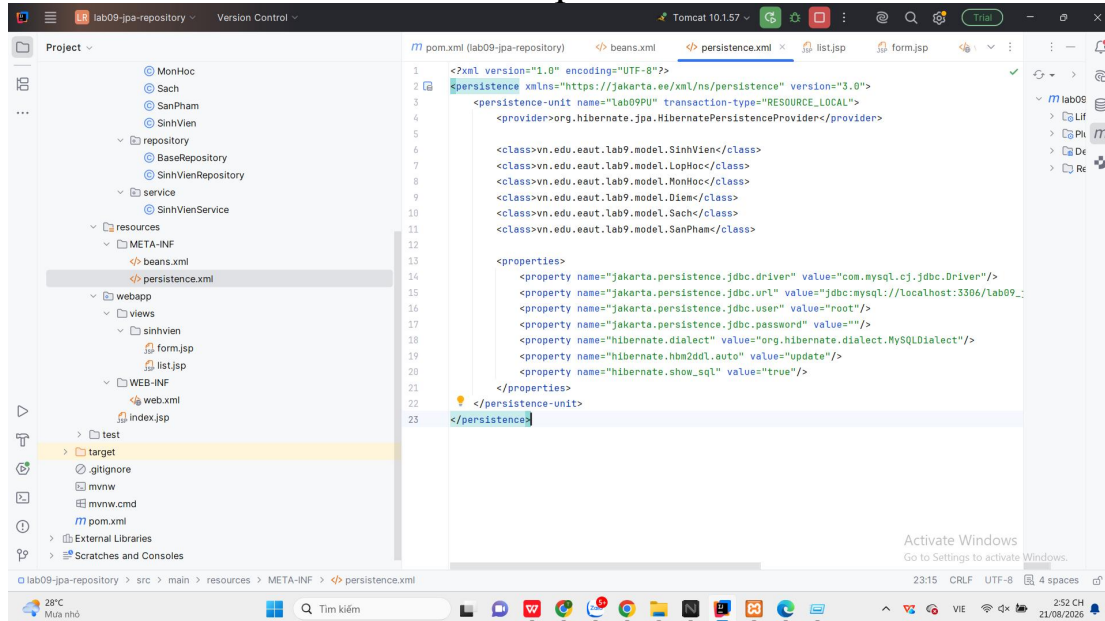


pom.xml



Bài 1. Cấu hình JPA và tạo lớp JPAUtil

src/main/resources/META-INF/persistence.xml



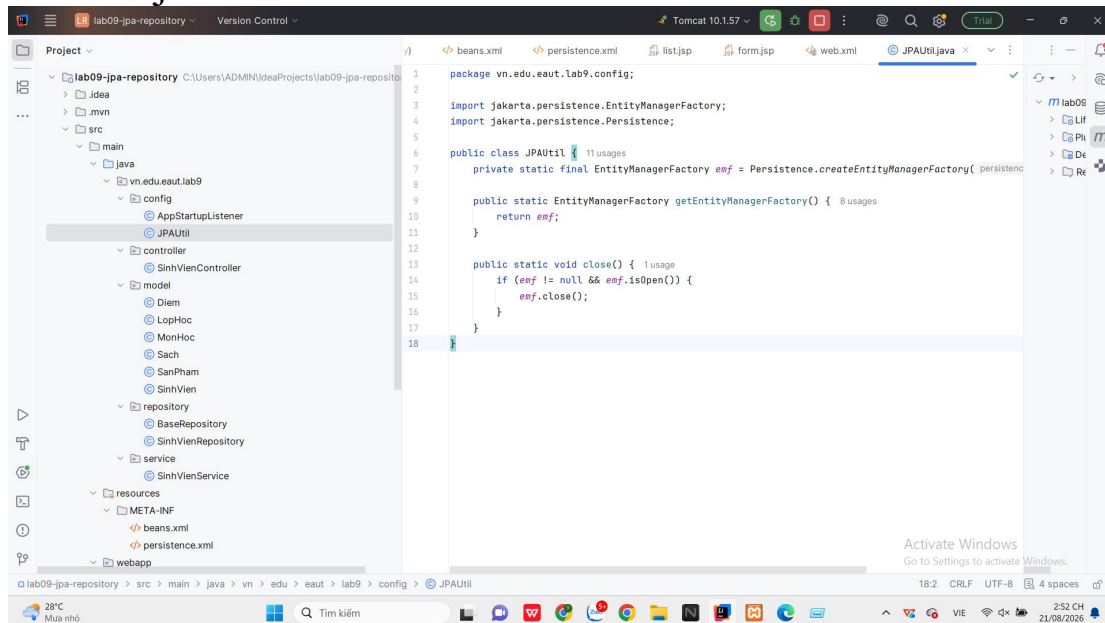
The screenshot shows an IDE window with the file `src/main/resources/META-INF/persistence.xml` open. The XML content is as follows:

```
<?xml version="1.0" encoding="UTF-8"?>
<persistence xmlns="https://jakarta.ee/xml/ns/persistence" version="3.0">
  <persistence-unit name="lab09PU" transaction-type="RESOURCE_LOCAL">
    <provider>org.hibernate.jpa.HibernatePersistenceProvider</provider>

    <class>vn.edu.eaut.lab9.model.SinhVien</class>
    <class>vn.edu.eaut.lab9.model.LopHoc</class>
    <class>vn.edu.eaut.lab9.model.MonHoc</class>
    <class>vn.edu.eaut.lab9.model.Diem</class>
    <class>vn.edu.eaut.lab9.model.Sach</class>
    <class>vn.edu.eaut.lab9.model.SanPham</class>

    <properties>
      <property name="jakarta.persistence.jdbc.driver" value="com.mysql.cj.jdbc.Driver"/>
      <property name="jakarta.persistence.jdbc.url" value="jdbc:mysql://localhost:3306/lab09_..."/>
      <property name="jakarta.persistence.jdbc.user" value="root"/>
      <property name="jakarta.persistence.jdbc.password" value="" />
      <property name="hibernate.dialect" value="org.hibernate.dialect.MySQLDialect"/>
      <property name="hibernate.hbm2ddl.auto" value="update"/>
      <property name="hibernate.show_sql" value="true"/>
    </properties>
  </persistence-unit>
</persistence>
```

JPAUtil.java



The screenshot shows an IDE window with the file `src/main/java/vn/edu/aut/lab9/config/JPAUtil.java` open. The Java code is as follows:

```
package vn.edu.eaut.lab9.config;

import jakarta.persistence.EntityManagerFactory;
import jakarta.persistence.Persistence;

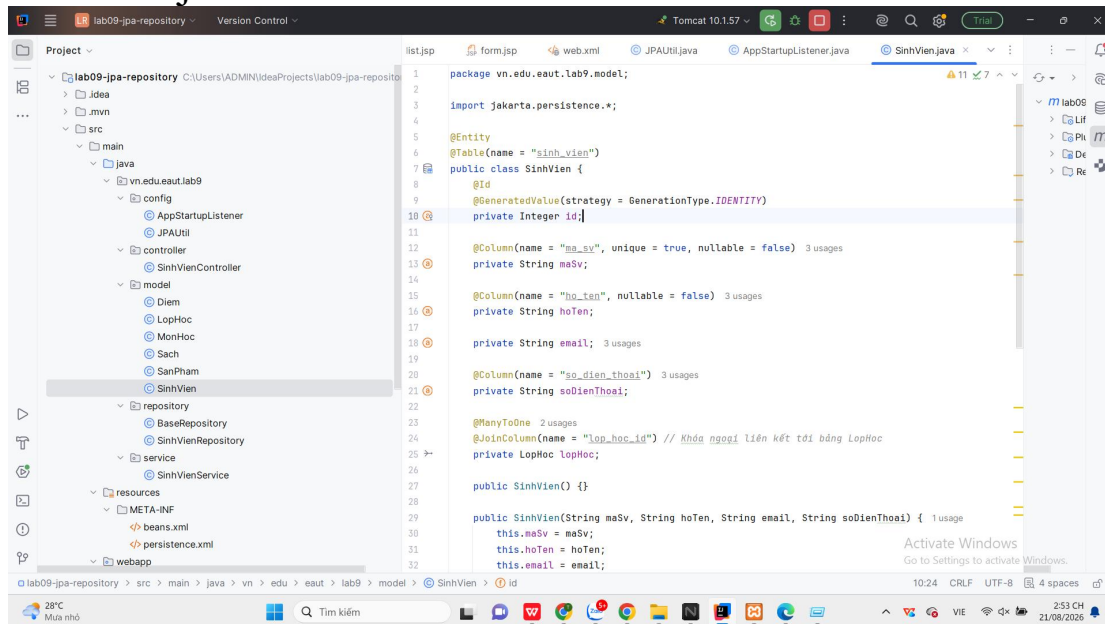
public class JPAUtil {
    private static final EntityManagerFactory enf = Persistence.createEntityManagerFactory("persistence-unit");

    public static EntityManagerFactory getEntityManagerFactory() {
        return enf;
    }

    public static void close() {
        if (enf != null && enf.isOpen()) {
            enf.close();
        }
    }
}
```

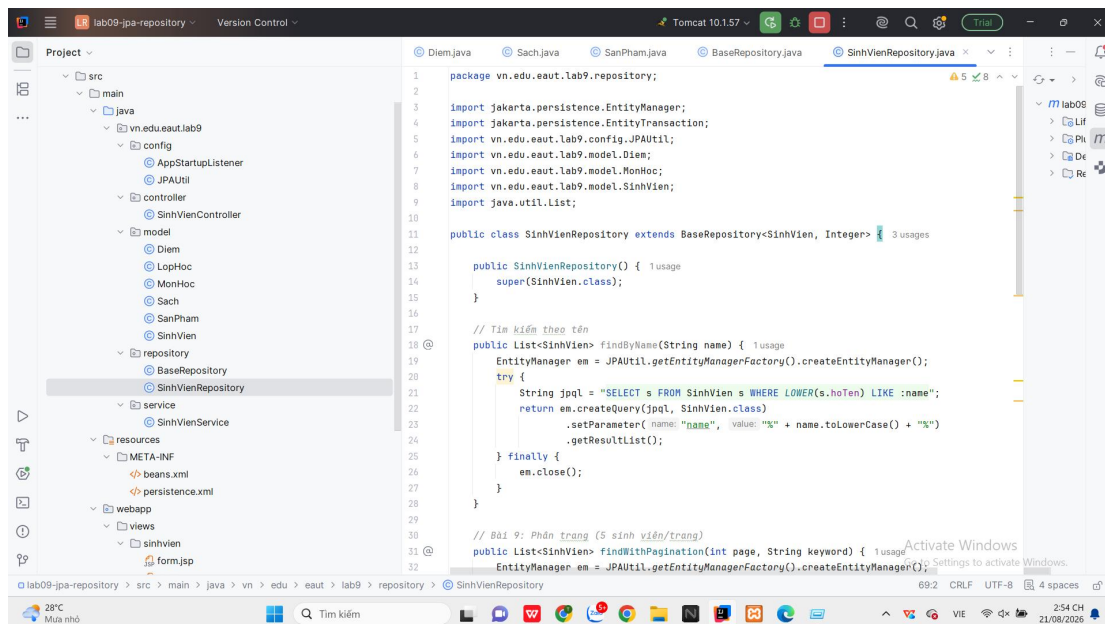
Bài 2. Tạo Entity SinhVien

SinhVien.java

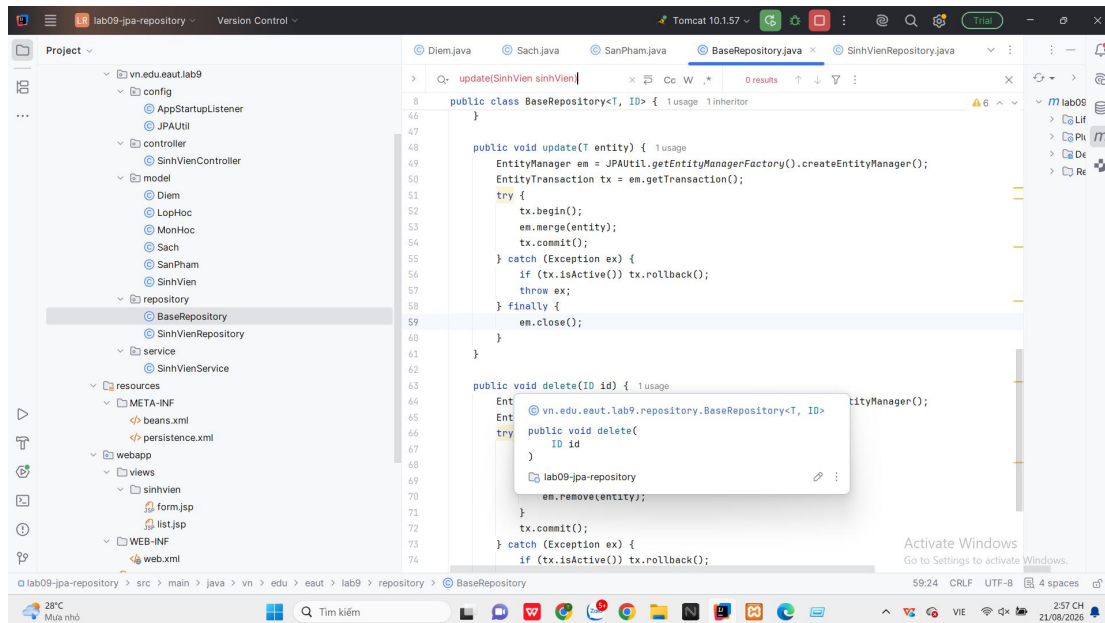


Bài 3. Xây dựng Repository CRUD bằng EntityManager

SinhVienRepository.java

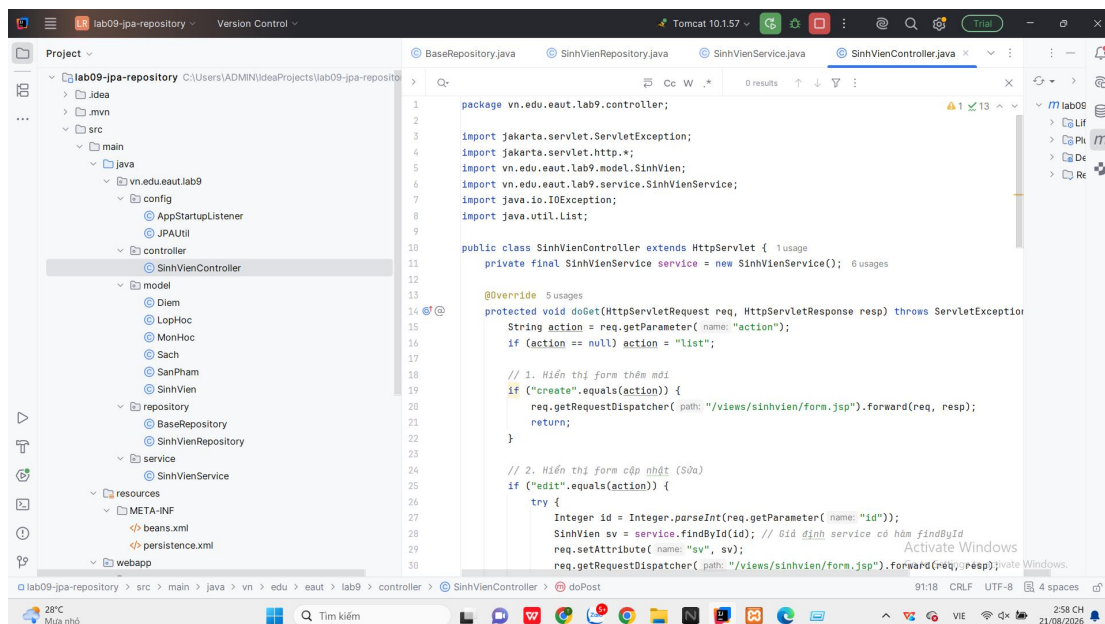


Bài 4. Bổ sung update, delete và tìm kiếm bằng JPQL

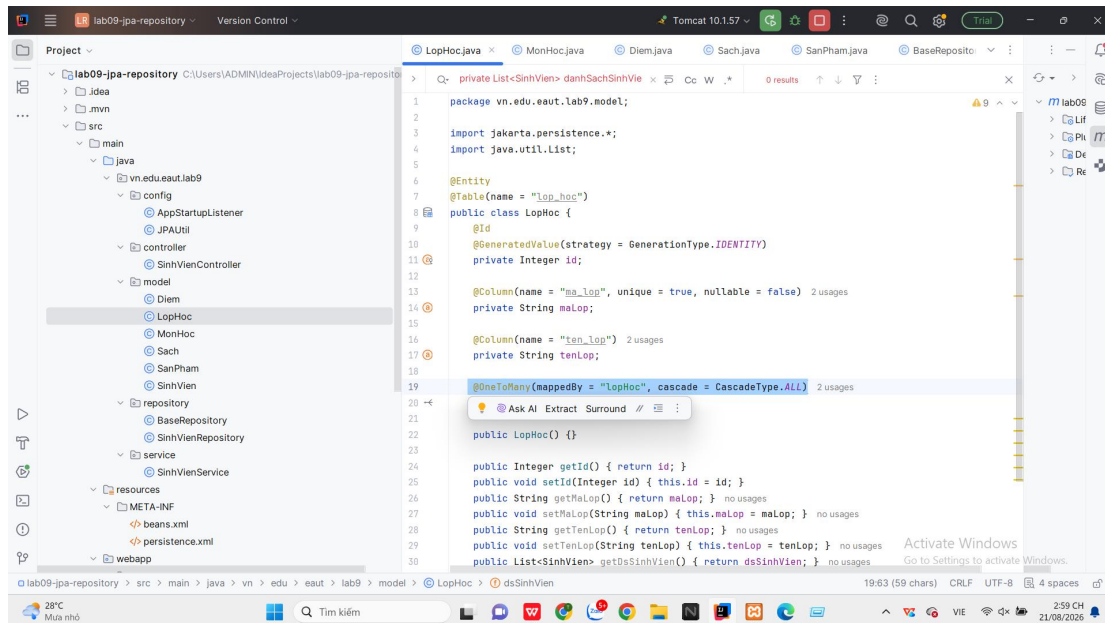


Bài 5. Kết nối Controller với Repository và hiển thị lên JSP/JSF

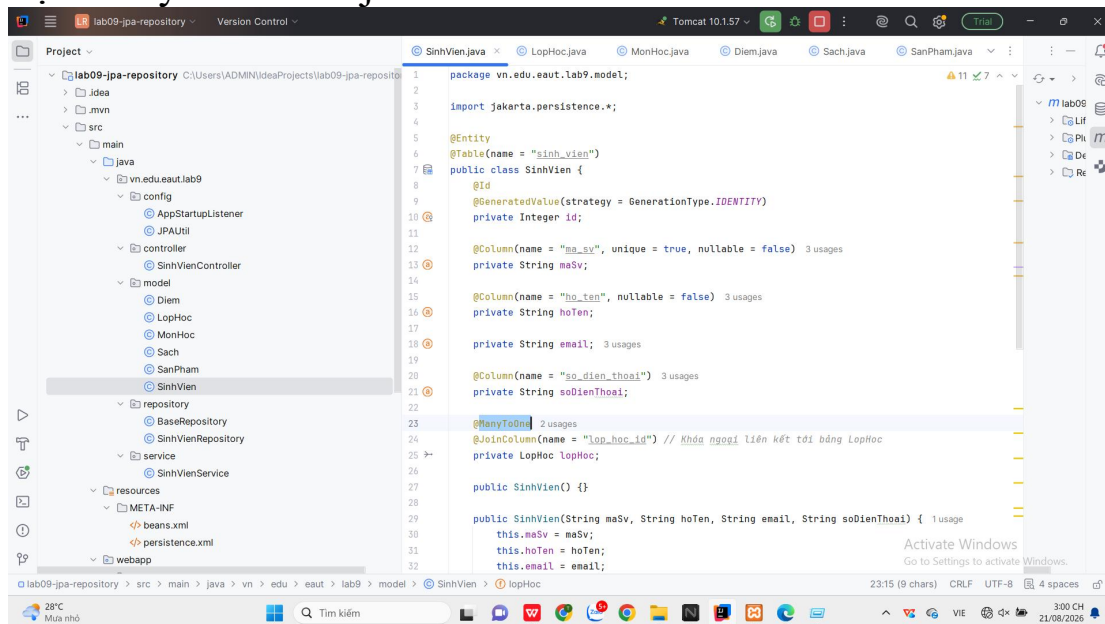
SinhVienController.java



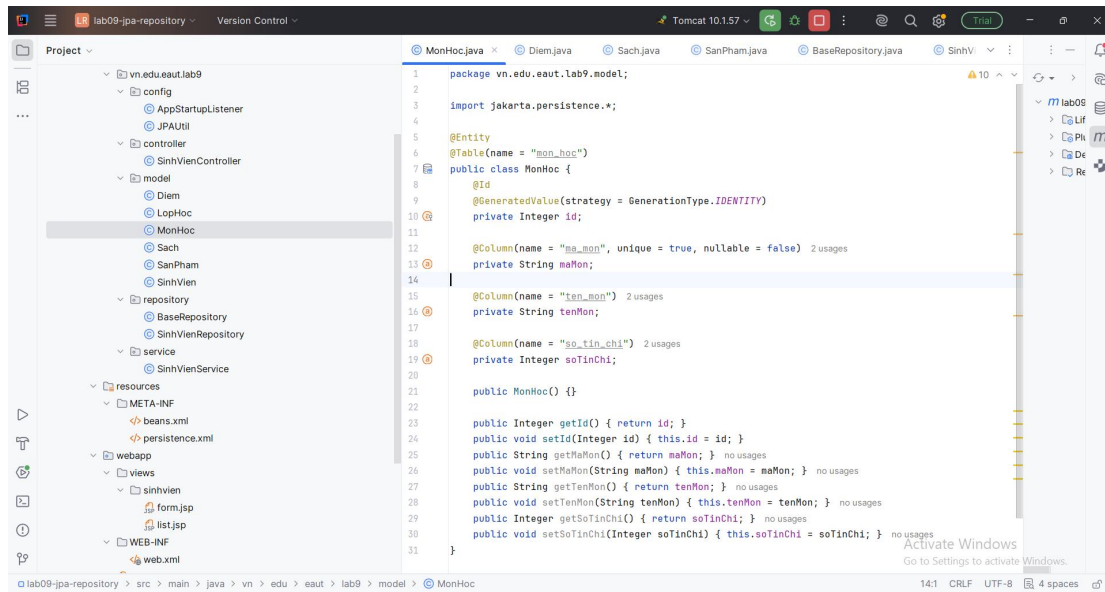
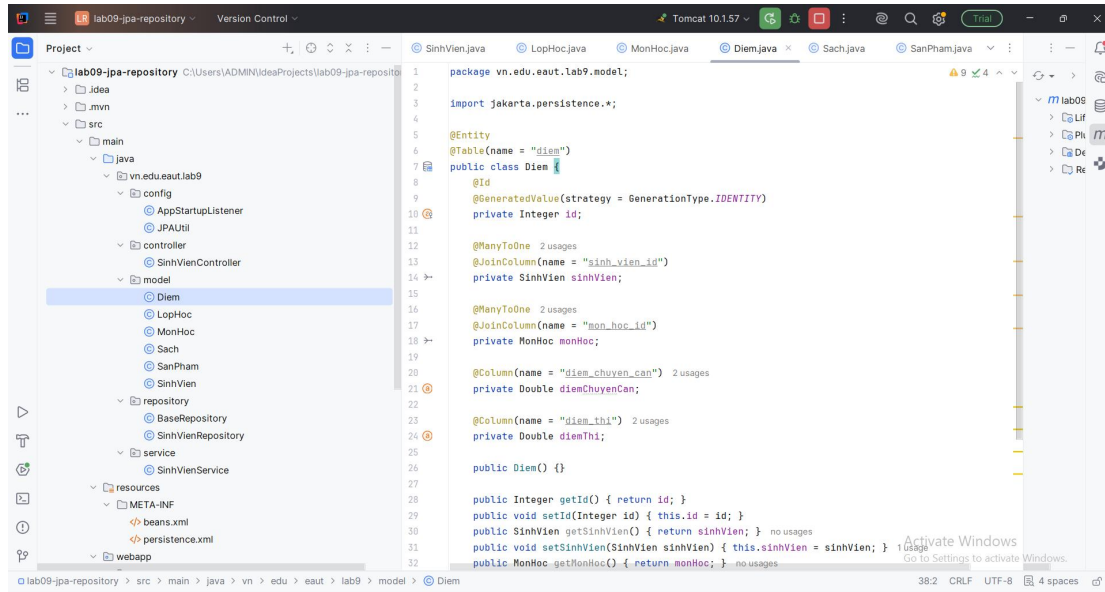
Bài 6. Entity Lớp học và quan hệ 1-nhiều



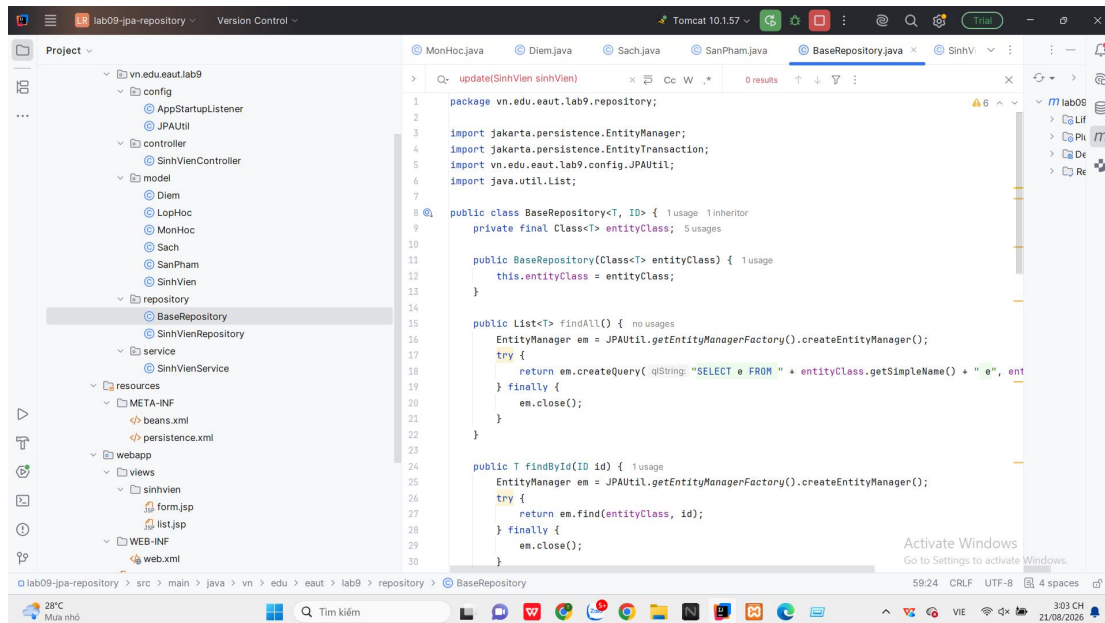
Tại Entity SinhVien.java



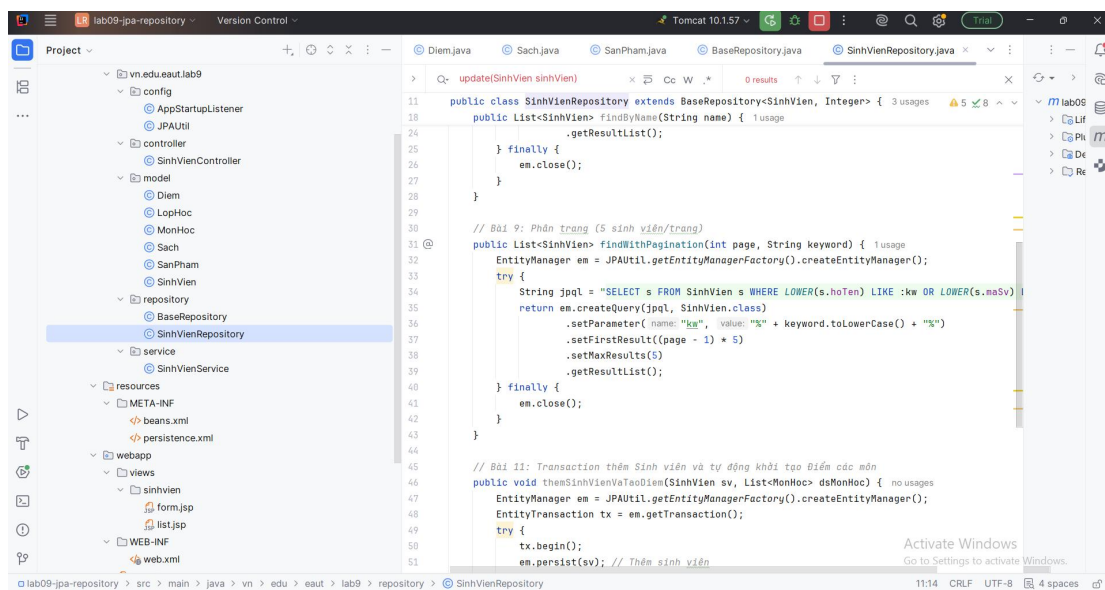
Bài 7. Entity Môn học và Điểm



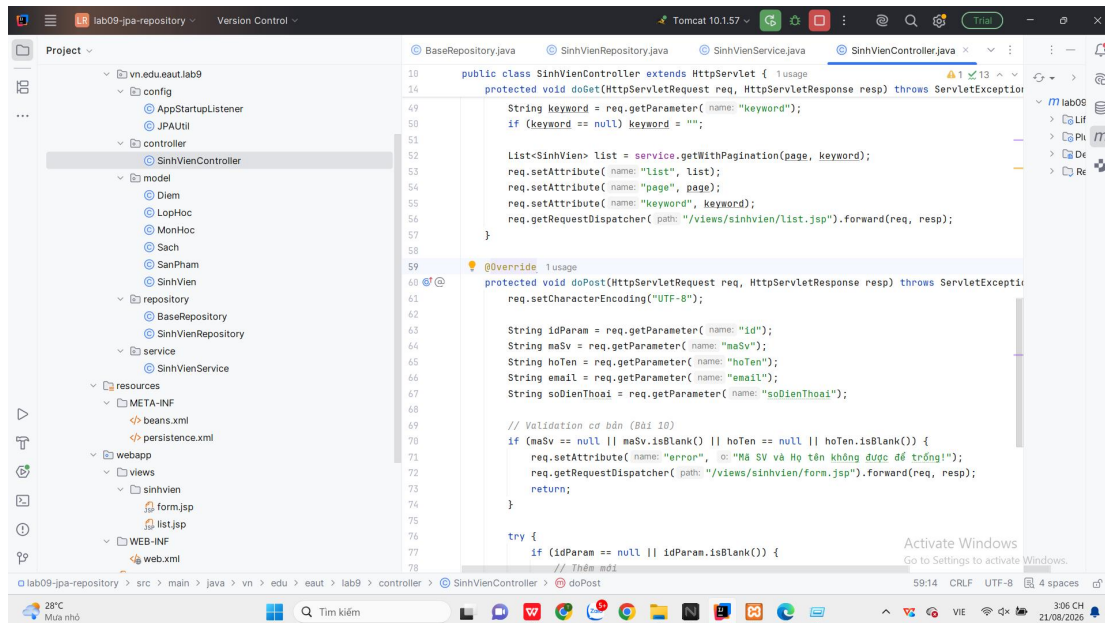
Bài 8. Repository dùng chung



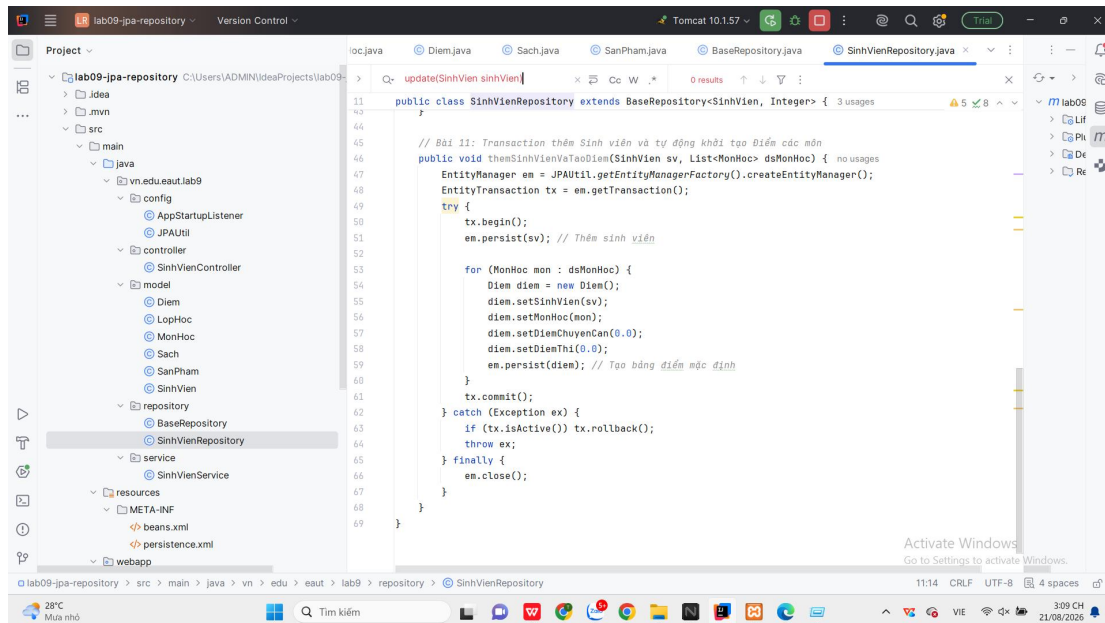
Bài 9 Phương thức phân trang và tìm kiếm trong Repository



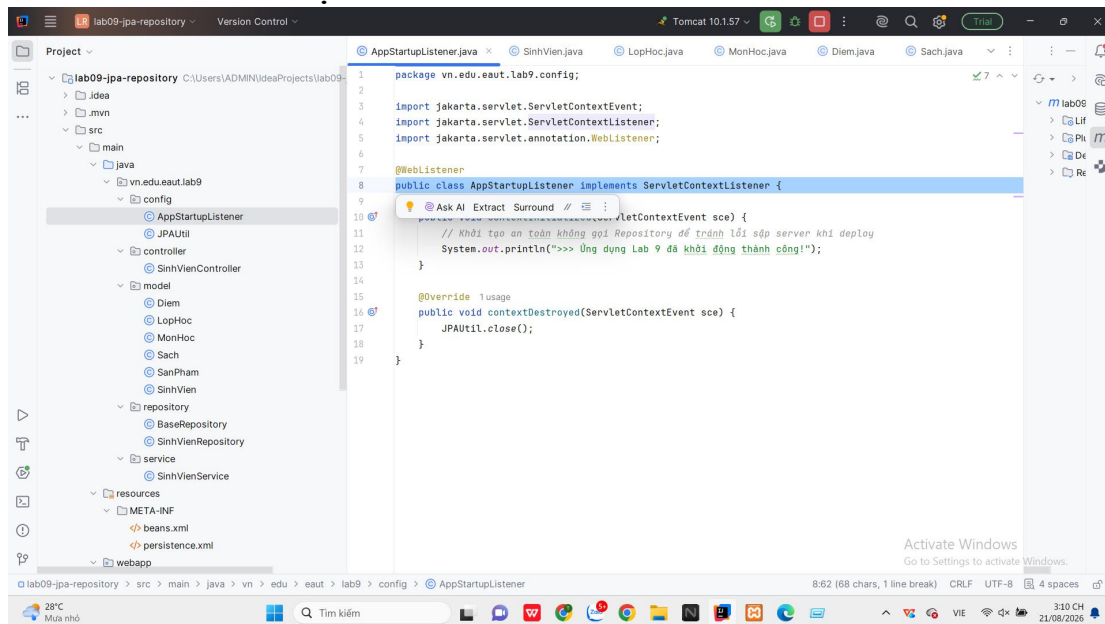
Bài 10. Ràng buộc dữ liệu và xử lý lỗi trùng mã



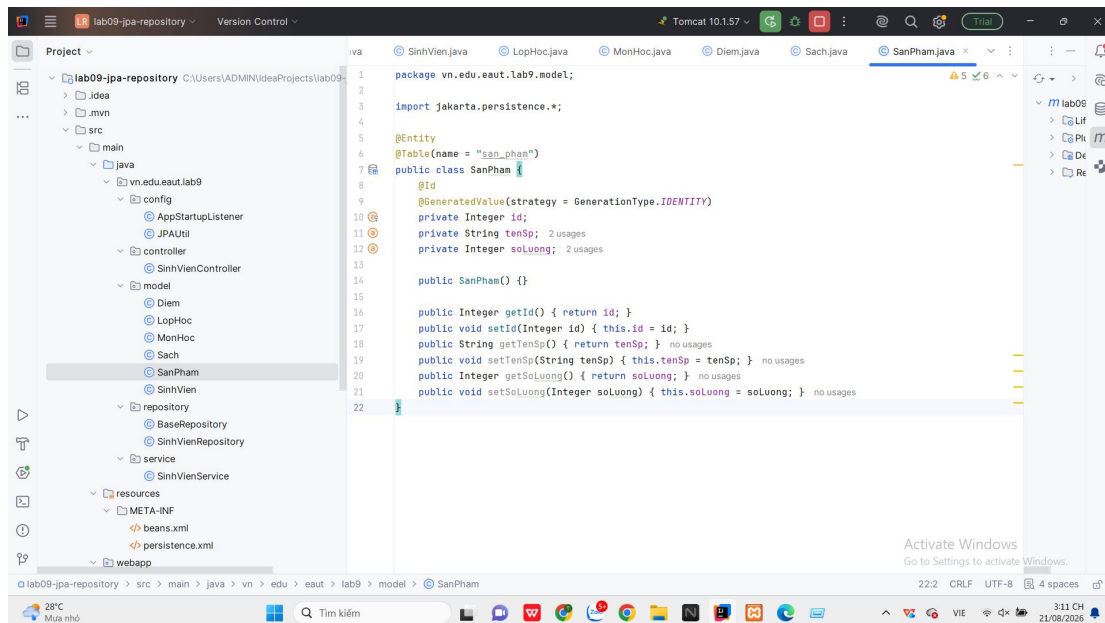
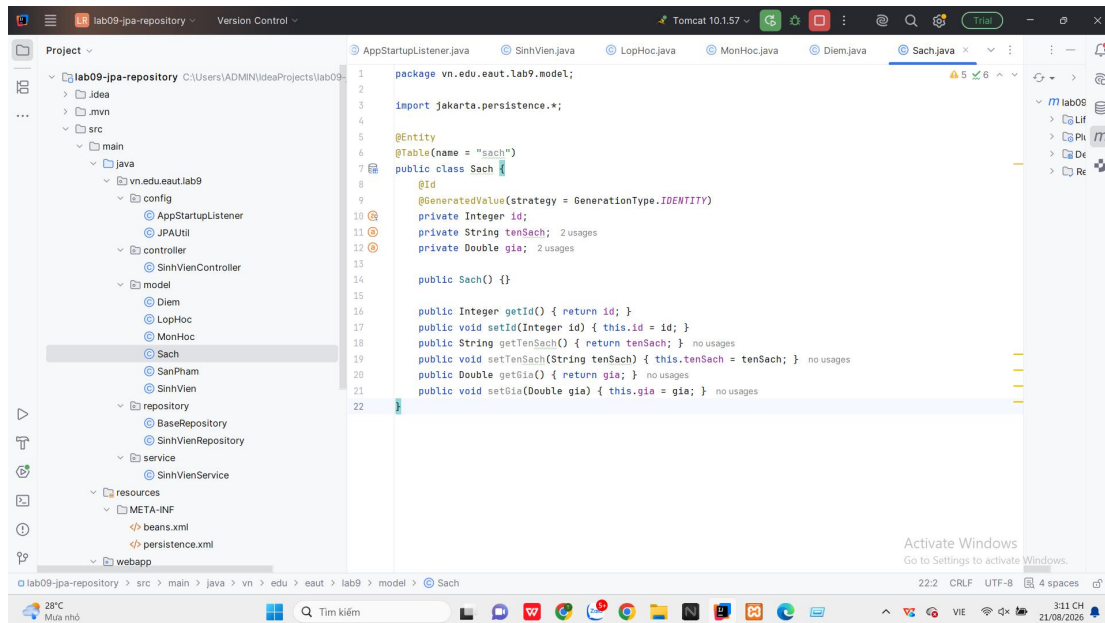
Bài 11. Transaction nhiều thao tác



Bài 12. Seed dữ liệu mẫu



Bài 13. Chuyển ít nhất 2 module từ List sang JPA



Chạy chương trình thành công

